
MapShift: Controlled Post-Intervention Evaluation for Embodied World Models

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Abstract

An agent can succeed in a world it explored yet fail to use that experience after the world changes. We introduce MapShift to measure this difference. In controlled post-intervention evaluation (CPE), reward-free exploration is followed by a matched, validated change to metric, topology, dynamics, or semantics, then by post-change tasks. An executable 24-motif reference benchmark reports outcomes by intervention family, task, and severity and checks whether each test is valid. Its mechanism study shows that unchanged-environment evaluation reduces the belief-update advantage from 0.304 to 0.102 for topology-family tasks and from 0.724 to 0.102 for semantic-family tasks. We also instantiate the protocol in held-out ProcTHOR houses. Across 371 contributing houses, local belief update beats stale-map reuse after target relocation by +0.532 success (95% CI [0.501, 0.562]), while their same-environment paired gap rounds to zero. With an RGB-input recurrent state-space model (RSSM) and fixed planner, grounded full repair gains +0.218 ([0.197, 0.242]) over grounded frozen belief under CPE. Finally, unchanged evaluation favors grounded full repair over RSSM-only full repair by 0.041, whereas CPE favors the latter by 0.224. Thus the evaluation protocol exposes update competence and reverses the choice between complete belief-and-planner systems under a controlled navigation interface.

1 Introduction

Suppose an agent explores a house and stores the location of a target. In the same house, both a system that reuses this memory and one that can revise it may navigate successfully. Move the target after exploration, and the two systems face a different test: can they use new local evidence to correct what they stored? Most embodied evaluations score behavior in a sampled environment without making this within-environment change explicit. An unchanged-world score therefore cannot isolate stale memory, post-change belief update, and planning with the updated belief. These are distinct competencies under partial observability [Kaelbling et al., 1998, Sutton and Barto, 2018].

Controlled post-intervention evaluation (CPE) makes this comparison explicit. An agent explores a base environment e without task reward for T_{exp} steps. A declared intervention $I_f(\sigma)$ changes one factor to form e' , and the agent then solves tasks in e' using its exploration-derived state and permitted post-change observations. The family f names the changed factor and σ its severity. Matched pairs keep the background world and task conditions fixed; validators check the realized edit and record rejected proposals. Running an unchanged-environment control on the same design shows which conclusions depend on the change itself. Figure 1 shows the sequence.

The paper develops the protocol in an inspectable 24-motif reference benchmark, then applies its matched-pair contract to semantic and topology edits in held-out ProcTHOR houses. Both studies matter. The reference study checks the measurement machinery and shows that same-environment evaluation compresses belief-update advantages and changes planning-slice rankings. In ProcTHOR, the same control hides a 0.532 local-update advantage after target relocation. Controlled systems built around an RGB-input RSSM separate under intervention and reverse their relative

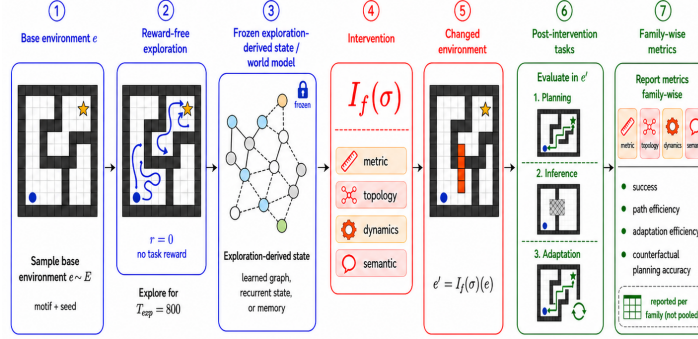


Figure 1: MapShift evaluates what an agent does with exploration-derived state after a declared change. Reward-free exploration in e precedes an intervention in one family at one severity; the agent then performs planning, inference, or adaptation tasks in matched e' . Validators check the realized change before results are reported by family, task, and severity.

order. The evaluated unit is the complete mapping from allowed exploration and post-change evidence to actions or predictions: an explicit model includes its belief interface and planner, while an end-to-end policy carries its own state through the same observation boundary.

Our contributions are (i) a protocol that holds the explored world fixed while declaring, validating, and measuring a post-exploration change; (ii) an executable reference benchmark with family-by-task-by-severity outputs, health gates, controls, and reproducible records; and (iii) evidence from both the reference study and certified ProcTHOR pairs that the evaluation choice changes what update mechanisms it can distinguish and which complete learned system it favors. The claims concern the stated navigation tasks and information interfaces.

2 Related work

Benchmarks for embodied agents. Reproducible agent benchmarks range from ALE, Gym, DeepMind Control/Lab, Procgen, and MiniGrid [Bellemare et al., 2013, Brockman et al., 2016, Tassa et al., 2018, Beattie et al., 2016, Cobbe et al., 2020, Chevalier-Boisvert et al., 2023] to open-ended Crafter and MineDojo [Hafner, 2022, Fan et al., 2022] and embodied simulators such as Habitat, AI2-THOR, ProcTHOR, VirtualHome, and ALFRED [Savva et al., 2019, Kolve et al., 2017, Deitke et al., 2022, Puig et al., 2018, Shridhar et al., 2020]. MapShift asks a different question across substrates: does knowledge from reward-free exploration of a particular world remain useful after a certified change to that world? Its ProcTHOR port validates realized edits, rejection reasons, and solvability in held-out indoor scenes.

World models and reward-free evaluation. World models learn predictive structure for planning and control [Ha and Schmidhuber, 2018, Hafner et al., 2020, Schrittwieser et al., 2020]; reward-free exploration separates information gathering from downstream reward optimization [Jin et al., 2020, Sekar et al., 2020, Mendonca et al., 2021]. WorldTest/AutumnBench likewise scores derived tests after reward-free interaction [Warrier et al., 2025]. MapShift specifically declares which factor changed after exploration, matches the before/after worlds, and compares protocols on the same generated study grid. This makes failure under changed topology or semantics interpretable as a family-conditioned outcome rather than only a low downstream score; Appendix A details the distinction.

Interventions, shift, and documentation. Causal RL studies interventions and counterfactual reasoning in sequential decisions [Lu et al., 2018, Buesing et al., 2019, Rezende et al., 2020]. MapShift uses known generator edits to diagnose robustness; it does not estimate causal effects from observational trajectories. Work on distribution shift and benchmark rankings shows that aggregates can conceal failures or change with the scoring rule [Koh et al., 2021, Maier-Hein et al., 2018, Perlitz et al., 2024, Chen et al., 2025]. We make family-by-task-by-severity results the default report and document edits, rejected proposals, solvability, and provenance, following calls for explicit evaluation scope [Geburu et al., 2021, Mitchell et al., 2019].

3 Controlled post-intervention protocol

Protocol. Let $\mathcal{E}$ be a distribution of environments, $\mathcal{F}$ the intervention families, and Σ_f the severity levels for family f . A system M explores $e \sim \mathcal{E}$ through observations and actions, without downstream task reward. Its resulting state is

$$z_M = \text{Explore}(M, e, T_{\text{exp}}). \quad (1)$$

Here z_M can be a recurrent state, explicit memory, learned graph, or the empty state in a no-exploration control. After exploration, $I_f(\sigma)$ changes one declared family $f \in \mathcal{F} = \{\text{metric, topology, dynamics, semantic}\}$ at level $\sigma \in \Sigma_f$:

$$e' = I_f(\sigma)(e). \quad (2)$$

A task $q \sim Q_c(e, e')$ comes from planning, inference, or adaptation class $c \in \mathcal{C}$. The system carries z_M into e' and is scored on its post-change response:

$$S(M; e, f, \sigma, c, q) = \text{Perf}(M, z_M, e', q). \quad (3)$$

The population target averages over environments, declared changes, and tasks:

$$\text{CPE}(M) = \mathbb{E}_{e \sim \mathcal{E}, f \sim \mathcal{F}, \sigma \sim \Sigma_f, c \sim \mathcal{C}, q \sim Q_c(e, e')} [S(M; e, f, \sigma, c, q)]. \quad (4)$$

The main estimand resolves that expectation by intervention family:

$$\text{CPE}_f(M) = \mathbb{E}_{e, \sigma \in \Sigma_f, c, q} [S(M; e, f, \sigma, c, q)]. \quad (5)$$

This separates failures of geometry, connectivity, dynamics, and semantic binding.

The default report further separates each family by task and severity, with absolute scores, paired effects, confidence intervals, and edit attrition. A mixed score is a secondary overview because planning, inference, and adaptation answer different questions. The unit being evaluated is the full state-construction and action interface, not only a predictive checkpoint.

Operational intervention semantics. We write the generator as $G(u, x)$. Here u fixes motif, seed, and layout randomness, while $x = (x_{\text{metric}}, x_{\text{topology}}, x_{\text{dynamics}}, x_{\text{semantic}})$ contains editable factors. A family intervention constructs

$$e' = G(u, x_{-f}, \phi_f(x_f, \sigma)), \quad (6)$$

where x_{-f} are the factors outside family f and ϕ_f is its transformation. Thus the pair shares the same background world; only the declared factor should change. Validators test that the intended edit occurred and non-target factors remained invariant. Episode records retain the realized edit, checks, and rejection provenance.

Empirical estimator. The executable study uses finite motifs, seeds, severities, tasks, and model seeds. Let $\mathcal{D}_f$ be valid episode records for M in family f after task rejection. The family score is

$$\widehat{\text{CPE}}_f(M) = \frac{1}{|\mathcal{D}_f|} \sum_{r \in \mathcal{D}_f} s_r, \quad (7)$$

where s_r is record r 's task-normalized score.

Primary score composition. For main family-primary tables, the primary score uses non-identity severities $\Sigma_f^* = \{1, 2, 3\}$ and equal weights over severity and task class:

$$\widehat{\text{CPE}}_f^*(M) = \frac{1}{3} \sum_{\sigma \in \Sigma_f^*} \frac{1}{3} \sum_{c \in \{\text{plan, infer, adapt}\}} \frac{1}{|\mathcal{D}_{f\sigma c}|} \sum_{r \in \mathcal{D}_{f\sigma c}} s_r. \quad (8)$$

Records within a family/severity/task cell have equal weight. Severity 0 checks identity behavior but is excluded from non-identity result tables such as Table 2. There, the protocol delta is $\Delta_{\text{protocol}}(f) = (\text{BU} - \text{stale})_{\text{CPE}, f} - (\text{BU} - \text{stale})_{\text{same}, f}$: how much more strongly CPE distinguishes belief update (BU) from stale reuse. The privileged reference bounds planning feasibility, not the mixed score across task classes.

Protocol comparison. Controls change one evaluation condition on the same generated grid. The same-environment control executes tasks in e instead of e' ; the no-exploration control empties z_M ; horizon controls shorten or lengthen task budgets. For protocol p , the induced method ordering is

$$\pi_p(f) = \text{rank}_M\left(\widehat{\text{CPE}}_{f,p}(M)\right), \quad (9)$$

We compare these orderings with Kendall τ , pairwise rank reversals, and changes in the top-ranked method or family-specific order.

4 Methodology

Executable study pipeline. Versioned configurations generate the study rather than hand-selected episodes. For each motif and seed, the pipeline writes a base world, identifier, and manifest. Systems explore for the configured budget; learned methods train or load checkpoints indexed by baseline, world, seed, and hyperparameters. The intervention and task generators then write matched worlds, tasks, invariant checks, rejection reasons, and episode records. A health report is produced before interpreting system scores.

Intervention construction. Table 1 specifies the intended change and preserved factors for each family. Level 0 is the identity sanity check; levels 1–3 increase the configured intervention magnitude. Severity is ordinal within a family, not a common difficulty scale across families. Validators test isolation and realized magnitude, while reference solvability and weak-baseline non-saturation test whether the resulting tasks are informative.

Table 1: Reference-benchmark interventions. Severity 0 is identity; the listed parameters rise through levels 1–3. Validators check both the target change and preserved factors.

Family	Severity parameter	Values	Preserved quantities
Metric	action-gain scale	1.0, 1.1, 1.25, 1.5	topology, semantic identities
Topology	blocker fraction	0.0, 0.08, 0.18, 0.32	semantic identities, action response
Dynamics	friction multiplier	1.0, 0.9, 0.75, 0.6	geometry, semantic identities
Semantic	remap fraction	0.0, 0.1, 0.25, 0.5	geometry, action response

Environment and observation model. The reference substrate has 96×96 occupancy maps at 0.25m resolution. Navigation uses a 0.25m forward primitive and 15° turns. Egocentric observations cover a 3.5m radius and 100° field of view, with semantic cues when visible. Each system explores from a navigable start in one episode; non-reference systems never receive privileged global state.

Exploration budget. All non-reference systems receive $T_{\text{exp}} = 800$ steps. Fixing this budget exposes how much useful knowledge a system extracts under partial coverage. The health report includes visited-cell and node ratios, path-length distributions, and weak-baseline saturation, so sparse coverage in larger motifs remains visible.

Worked example. In a topology episode, the agent explores a two-room world and remembers a traversable corridor. Blockers are then inserted, with cue identities and dynamics preserved. A stale planner still routes through the corridor; an updater can revise its map from local post-change observations. The unchanged control scores execution in the old world, whereas CPE scores the same system after the corridor change.

Intervention validation. Validators check what actually changed: metric edits must preserve topology and semantics; topology edits preserve cue identity and action response; dynamics edits preserve geometry; semantic edits preserve navigable geometry and reachability. These evaluator-side checks are hidden from systems. A failed check is reported before results and excludes the invalid pair.

Task sampling. Comparable conditions share start, goal, visibility, and horizon policies. Planning tests navigation, rerouting, changed dynamics, and cues; inference tests topology detection, masked-state prediction, and counterfactual reachability; adaptation allows limited post-change interaction

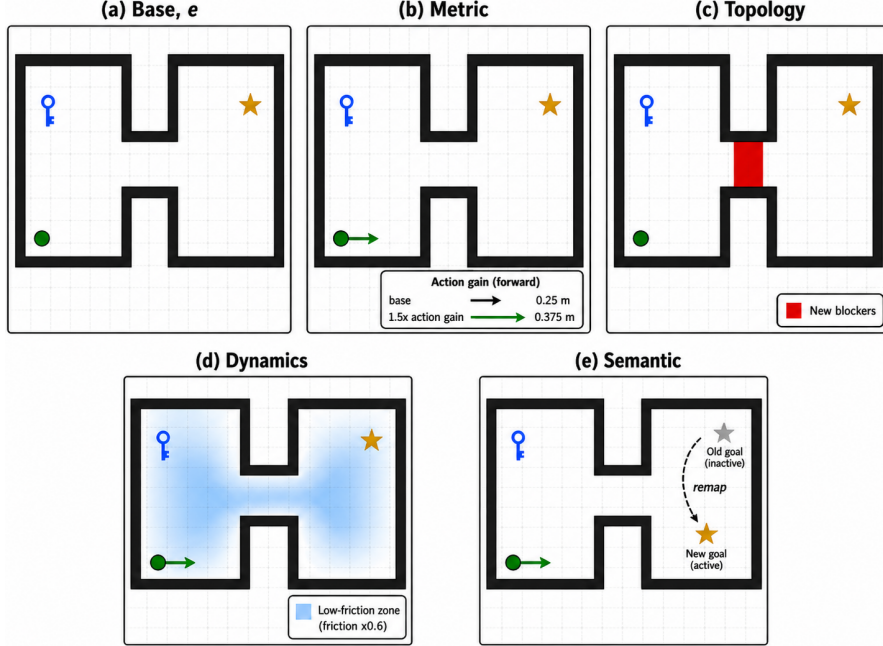


Figure 2: Matched reference-world pairs. The same two-room world (a) receives a level-3 metric rescaling (b), corridor blockers (c), low-friction dynamics (d), or goal-token remap (e). Motif, seed, key location, and start remain fixed; validators check the declared edit and non-target invariants. The study also includes severity levels 0–2.

before replanning. Planning horizons are 64/96/128 steps, inference horizons 16/32, and adaptation budgets and horizons 16/32/64.

Task rejection and solvability. The sampler rejects infeasible tasks, vacuous queries whose answer does not reflect the intended change, and trivial paths or horizons. It records reasons by family/severity/task cell. Rejection accounting distinguishes difficult valid cells from cells uncovered because valid tasks could not be generated.

Baseline evaluation. The privileged planner sees the changed world solely to check solvability and calibrate planning. A same-environment control executes tasks in the explored world. The stale-map planner stores the pre-change map without revision; classical belief update revises occupancy and token bindings from local observations without an intervention label, mismatch location, or global post-change map. Pretrained graph, persistent-memory, relational-graph, and structured-dynamics systems use fixed hyperparameters as learned calibration probes. Records identify the system, seed, protocol, motif, split, family, severity, task, and bootstrap group; Appendix B details the baselines.

Weak heuristic floor. The weak heuristic stores visited cells, visible token bindings, and shallow local traversability. It plans within remembered local structure without global repair, providing a nontrivial floor that should not saturate post-change tasks. Appendix B specifies its behavior.

Information access. Non-reference methods receive their exploration-derived state, task specification, and permitted local post-change observations. They receive neither the global post-change map nor the intervention label.

Scoring. Planning records include success, executed and reference path lengths, and post-change correctness. Path efficiency for record r is

$$\eta_r = \begin{cases} \min(\ell_r^*/\ell_r, 1) & \text{if the agent succeeds and } \ell_r > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (10)$$

Here ℓ_r^* is the shortest feasible path in the evaluation world and ℓ_r the executed path. Failures score zero. Post-intervention planning accuracy uses tasks whose answer depends on the edit, excluding

identity sanity cases. Reference-path quantities remain separate from the mixed family score; a privileged planning reference is not an upper bound on that score across task types.

Inference uses exact-match accuracy for masked-state and post-change reachability queries; topology detection uses average-rank AUROC with tied ranks. Adaptation records a recovery curve over limited post-change interaction and summarizes sample efficiency. The reference study reports 95% intervals from 1,000 resamples grouped by environment and model seed, keeping repeated tasks in a group together. Section 7 specifies the house-level grouping for ProcTHOR.

5 MapShift benchmark design

The reference benchmark is a partially observed navigation world with occupancy, transition, and semantic layers. These layers expose geometry, connectivity, dynamics, and cue meaning separately while keeping interventions and validators inspectable. The release covers 24 motifs, four families, severities 0–3, three task classes, three samples per class/cell, and an 800-step exploration budget. Splits are by motif, with eight held-out test motifs; full values appear in Appendix A. The reference world makes the measurement machinery easy to inspect, while the ProcTHOR study tests the contract in held-out 3D scenes.

6 Benchmark health and validity checks

MapShift checks the test before scoring systems. Its health report covers split separation, coverage by family/severity/task, rejection reasons, reference solvability, weak-baseline saturation, paths and horizons, realized intervention magnitude, isolation, and leakage.

Final-release task rejection is counted after internal proposal resampling. Thus zero rejections means valid, non-vacuous released tasks cover every cell; it does not mean every proposed task passed. The severity gate checks that configured change magnitude rises from level 0 to 3, not that finite-sample model scores are monotone. Weak-baseline saturation or flat success/failure across levels triggers a health warning.

Fatal split or task leakage invalidates a result; diagnostic warnings prompt inspection; reusable templates are labeled when they reveal no held-out world. In the completed reference run, fatal leakage, warnings, intervention-validator failures, severity-magnitude failures, and final-release task rejections are all zero. Reference solvability is 1.000; Table A.2 lists the health values.

7 Evaluation design

Baselines and role. The reference study compares a privileged planning diagnostic, weak heuristic, stale-map planner, classical belief updater, and learned graph, memory, and dynamics probes. The learned rows test capacity and representation choices; they are calibration probes rather than a leaderboard of contemporary agents. The primary mechanism comparison pairs stale reuse with explicit local update on the same generated grid. The ProcTHOR study then tests analogous planning schedules and controlled interfaces around a trained RGB-input RSSM.

Study grid and aggregation. Reference learned baselines use five seeds; larger pretrained-graph and persistent-memory rows cover 2,592 non-identity episodes. Versioned configurations fix hyperparameters before final evaluation, and validation motifs serve pre-final choices. The mechanism study compares CPE with same-environment and no-exploration controls on the same grid, reporting paired update-minus-stale gaps and ranking changes. Resampling retains repeated tasks within environment/model-seed groups.

Held-out 3D study and uncertainty. We screen 448 official ProcTHOR test-split houses. A semantic pair holds house, start pose, target identity, and rendering fixed while relocating the target within a declared navigable-geometry tolerance; a topology pair closes an interior door. Certification checks the edit as realized in the simulator, including geometry, feasibility, and non-target coupling. Of 43,633 candidate edits, 5,523 pass across the two families; the rejection ledger retains reasons for excluded proposals. The semantic navigation analysis uses 371 contributing houses. Systems

see only observations within 1.5 m, without the intervention label, mismatch location, or global post-change map.

The RSSM is trained on 3,260 official training houses and evaluated over three seeds; no test reachable-lattice hash matches a training hash. Within each seed, evaluated variants share checkpoint, planner, exploration trace, initial recurrent state, task instances, observation interface, and action budget. We pool seeds within house and average house means. Paired 95% confidence intervals resample houses, retaining all compared systems and seeds from a sampled house together. This design makes the house, rather than repeated episodes in it, the uncertainty unit.

The artifact README gives the smoke-test, health-audit, result-reproduction, and MiniGrid-adaptor commands and expected outputs; Appendix F summarizes the paths.

8 Results: what the protocol distinguishes

The two studies test complementary parts of the same measurement claim. The inspectable 24-motif release verifies intervention validity and shows which mechanisms an unchanged-world control hides. The held-out ProcTHOR study asks whether the distinction remains useful for local updates and learned systems in 3D scenes. We report the reference mechanism diagnostic first, then the matched-house navigation study.

Reference mechanism diagnostic. All reference health gates pass: zero final task rejections, fatal leakage, leakage warnings, intervention-validator failures, and severity-magnitude failures; reference solvability is 1.000 and the weak-heuristic health success proxy is 0.415. Across all 24 motifs, classical belief update exceeds stale-map reuse when topology or semantic bindings change (0.689 versus 0.395 and 0.810 versus 0.178, respectively). Stale reuse remains competitive on dynamics shifts, where geometry is preserved, matching the planning reference at 0.656. This family separation is the point of the diagnostic.

On eight held-out motifs, unchanged-environment evaluation compresses the belief-update advantage by roughly threefold for topology (0.304 under CPE versus 0.102 unchanged) and sevenfold for semantics (0.724 versus 0.102). The paired protocol deltas in Table 2 are positive for both families. The result identifies what the evaluation measures: a system that never revises its stored world can look much closer to an updaters if the test world never changes.

Table 2: Reference mechanism diagnostic on eight held-out motifs. Gaps are belief update minus stale-map score over non-identity severities 1–3; the protocol delta subtracts the unchanged-world gap from the CPE gap. Brackets show paired grouped-bootstrap 95% intervals.

Family	CPE gap	Same-env. gap	Protocol delta
Topology	0.304 [0.189, 0.424]	0.102 [0.082, 0.125]	0.201 [0.095, 0.306]
Semantic	0.724 [0.692, 0.749]	0.102 [0.080, 0.120]	0.622 [0.611, 0.631]

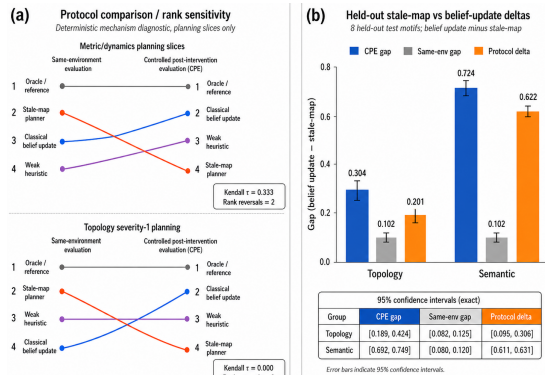


Figure 3: The reference protocol changes the conclusion. Panel (a) contrasts unchanged-world and CPE planning-slice rankings; panel (b) shows that held-out update-minus-stale gaps widen from 0.102 to 0.304 on topology and from 0.102 to 0.724 on semantic shifts.

The conclusion also depends on the task slice. In planning, unchanged-world and CPE orderings have Kendall $\tau = 0.333$ with two reversals on metric/dynamics tasks and $\tau = 0.000$ with three on topology severity 1. The mixed family score combines planning, inference, and adaptation and can retain its overall order (Table A.4) even when these planning conclusions change. That is why the default report preserves the slices.

Belief update wins on every held-out topology and semantic motif under CPE. The protocol enlarges its advantage on seven of eight topology motifs and all eight semantic motifs (Table A.5). That motif-wise descriptive summary is separate from the primary paired grouped estimate in Table 2.

Held-out 3D navigation. The ProcTHOR pairs retain the same house and start while changing a target location or closing a door. Figure 4 shows two semantic-navigation conclusions: local updating becomes useful only after target relocation, and the preferred learned-system belief interface reverses. Table 3 gives the five controlled RSSM variants. Navigation success is primary; path efficiency uses the post-change reference path and scores failures as zero.

Local update in ProcTHOR. After target relocation, stale-map reuse succeeds in 0.109 of episodes and local belief update in 0.641: a paired $+0.532$ success gap (95% CI $[0.501, 0.562]$) across 371 houses. In unchanged houses the paired gap rounds to 0.000; separately rounded arm means are 0.796 and 0.797. Path efficiency likewise gains $+0.239$ $[0.219, 0.259]$ after relocation and 0.000 unchanged. The updater uses local post-change evidence; an instrumentation check found no out-of-radius planning cells in 369 audited episodes. Without exploration memory, its success falls from 0.641 to 0.183, tying the gain to stored experience and local revision. For door-closure topology edits, reactive rerouting weakens binary-success discrimination, but path efficiency still favors update by $+0.056$ $[0.040, 0.072]$.

Learned-system discrimination. An RSSM trained from egocentric RGB and pose decodes occupancy and target beliefs for a fixed BFS planner; held-out occupancy IoU is 0.56–0.58. Five systems share the learned checkpoint and evaluation conditions within each seed. A grounded belief supplies observed cells to the planner; an RSSM-only belief reconstructs the planning map from recurrent state. Repair can refresh occupancy every eight steps, rebind a refuted target, or combine both. These are changes to the complete belief-and-planner interface, not separately trained model architectures.

The four grounded variants all score 0.859 to displayed precision without intervention, yet span 0.115–0.333 after relocation. Grounded full repair beats grounded frozen belief by $+0.218$ $[0.197, 0.242]$ success and $+0.143$ $[0.128, 0.160]$ path efficiency under CPE. Occupancy repair alone reaches 0.116, target rebinding 0.225, and both together 0.333. The combined $+0.218$ gain exceeds the sum of the isolated gains (about $+0.11$), revealing an interaction between free-space repair and target rebinding that unchanged evaluation cannot distinguish. Table 3 shows the absolute success values and intervals.

Table 3: Semantic navigation success in 371 held-out houses (three RSSM seeds). Within seed, systems share checkpoint, planner, episodes, and budget; A and B use full repair but differ in planning-belief construction. Brackets are 95% house-bootstrap intervals.

Evaluated system	Same environment	Semantic CPE
Grounded, frozen belief	0.859 $[0.837, 0.881]$	0.115 $[0.096, 0.135]$
Grounded, occupancy repair	0.859 $[0.836, 0.881]$	0.116 $[0.098, 0.135]$
Grounded, target-binding repair	0.859 $[0.836, 0.881]$	0.225 $[0.199, 0.251]$
Grounded, full repair (A)	0.859 $[0.837, 0.881]$	0.333 $[0.306, 0.360]$
RSSM-only, full repair (B)	0.818 $[0.799, 0.838]$	0.557 $[0.527, 0.588]$

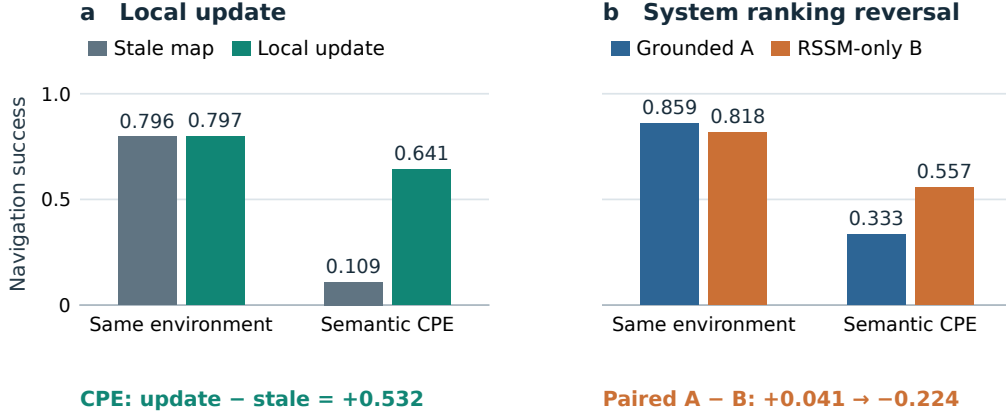


Figure 4: Two conclusions revealed by semantic CPE in ProcTHOR. (a) Local update and stale reuse tie in the unchanged-world paired contrast, but update gains +0.532 [0.501, 0.562] success after target relocation. (b) The preferred full-repair system reverses: unchanged evaluation favors grounded belief (A) over RSSM-only belief (B) by +0.041 [0.017, 0.065]; CPE favors B by +0.224 [0.192, 0.255]. A and B share checkpoint and planner within seed.

The reversal changes which belief interface would be selected: direct grounding helps in unchanged houses, whereas reconstructing the map from updated recurrent state wins after target relocation. Checkpoint, planner, repair schedule, and information boundary are paired within seed. The inference concerns these complete systems under this navigation interface; it does not rank independently trained architectures.

Sensitivity and reporting. The ProcTHOR semantic update gap remains positive at 200, 400, 800, and 1600 exploration steps (+0.267, +0.400, +0.532, and +0.614), while unchanged-world gaps stay at most 0.002. We use 800 steps before unchanged-world success nears saturation at longer budgets. The semantic conclusion also stays positive across 231 tested severity weightings of certified relocations and 2,000 random house reweightings. Across studies, family $\times$ task $\times$ severity scores, paired intervals, and rejection counts are the default output; mixed scores remain a secondary overview.

9 Interpreting and using the evaluation

What the controls identify. The reference and ProcTHOR comparisons are paired tests of the evaluation condition, not just demonstrations that a stronger updater can outperform a stale planner. In the reference study, the same generated worlds and mechanisms are scored under CPE and unchanged-world controls; the update advantage grows when the stored topology or semantics become wrong. In ProcTHOR, the house, start, target identity, and rendering are paired while the target is relocated. The unchanged-house control collapses the local-update gap, whereas the post-change test separates it. A no-exploration or memory-removal control further asks whether the advantage depends on what the agent learned before the change. These contrasts distinguish the value of stored experience, its revision, and subsequent planning under a stated observation boundary.

Why certification is part of the result. In a generated grid, the intended edit can be checked against explicit layers. In a 3D simulator, moving an object or closing a door can also change visibility, navigability, or feasibility. The evaluator therefore certifies the realized pair and records proposals that fail the declared tolerance; 5,523 of 43,633 ProcTHOR candidates pass across the two tested families. An informative score requires both a valid changed world and an agent that was not shown evaluator-side metadata. The 371-house semantic analysis and house-clustered intervals then describe variation across independent scenes, rather than treating many episodes from one house as independent evidence. This contract can be ported only where the intended edit and its side effects can be checked.

What a system comparison means. The RSSM result does not isolate a network architecture from the rest of the agent. It holds checkpoint and planner fixed within seed while changing how observed cells and recurrent state form the planning belief and how that belief is repaired. The four grounded variants tie without intervention but differ after relocation; two full-repair variants even exchange order. For an explicit predictive model, a benchmark should therefore specify the route from observations through belief to action. An end-to-end policy can be scored under the same exploration, intervention, observation, and action boundary while carrying its own internal state. In either case, absolute family/task/severity outcomes and paired protocol effects convey more than a single mixed leaderboard value.

Failure mode and metric must be named together. The interventions do not merely make every task harder. In the reference study, stale geometry is especially damaging for topology and semantics, while a stale planner can remain competitive on dynamics when geometry stays valid. In ProcTHOR, semantic target relocation produces a large success separation, but reactive rerouting after door closure can make topology success less informative even when path efficiency differs. An aggregate score would blur both observations. Reporting the family, task, severity, and metric alongside a paired effect tells a reader whether the claim concerns reaching a goal, taking an efficient route, answering a changed-world query, or adapting with limited new evidence.

Minimum reporting contract. A new instantiation should publish the base/change pair identifiers, declared edit and tolerance, realized validator outcomes, and the ledger of rejected proposals before presenting model scores. It should state which observations and intervention metadata are visible to the agent, verify reference solvability, and document coverage and weak-baseline saturation by cell. The same starts, tasks, and seeds should be paired across protocol controls where feasible; intervals should resample independent worlds rather than episodes within one world. Absolute scores and paired differences belong in the family $\times$ task $\times$ severity report, with a mixed score only as a summary. These records let others tell an informative protocol effect from task attrition, accidental access to the edit, or a choice of aggregation.

10 Limitations and future work

The 24-motif reference world favors inspectable edits over visual complexity. ProcTHOR extends the contract to certified target relocations and door closures in held-out 3D navigation scenes; these results do not establish transfer to manipulation, language grounding, contact dynamics, or robots. The RSSM study pairs checkpoints and a fixed planner to isolate belief interfaces, rather than rank independently trained agents. Reactive rerouting can saturate topology success, so path efficiency remains important. A new simulator or hardware setting needs its own checks of realized coupling, attrition, solvability, and observation access. Our claims are limited to the reported tasks, systems, and aggregation choices.

11 Conclusion

MapShift measures whether exploration-derived state remains useful after a declared, validated change to the same world. In the 24-motif reference study, unchanged evaluation compresses belief-update gaps and changes planning-slice rankings. In 371 held-out ProcTHOR houses, target relocation exposes a 0.532 local-update advantage hidden by the unchanged control, separates repair choices around one RSSM, and reverses the preferred full-repair belief interface. Matched edits, explicit information boundaries, and family-resolved reports make these conclusions auditable. The two studies show why post-change tests can change what we infer from an embodied world model’s score.

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Appendix

A Additional result tables

Table A.1: Conceptual distinction from reward-free derived-test benchmarks such as WorldTest/AutumnBench. The comparison is about what claims the protocol can identify, not which simulator is more realistic.

Axis	Derived-test protocol	MapShift CPE protocol
Changed factor	Scores a downstream test after reward-free interaction; the changed factor need not be declared as an estimand.	Declares metric, topology, dynamics, or semantic intervention family and severity.
Matched pair	May compare performance across derived tasks or generated worlds.	Holds motif, seed, start state, and non-target factors fixed across e and e' .
Failure attribution	A low score can indicate weak memory, weak update, weak planning, or task mismatch.	Family-wise estimands localize failures to declared environment factors.
Protocol ablations	Not necessarily tied to a same generated intervention grid.	Same-environment, no-exploration, and horizon controls are run on the same grid.
Validity checks	Depends on benchmark-specific documentation.	Emits intervention validators, leakage checks, rejection counts, solvability, and saturation gates.

Table A.2: Benchmark health summary for the expanded release preflight. Health checks are generated before model-result interpretation and included in the release artifact.

Check	Value
Environments and splits	24 motifs; 10 train, 6 validation, 8 test
Task coverage	3456 tasks; 0 task rejections
Reference solvability	1.000
Weak-baseline health estimate	0.415 success proxy; not primary CPE score
Fatal leakage / benign warnings	0 / 0
Intervention validator failures	0
Severity-magnitude failures	0

Table A.3: MapShift release values. These values define the main study and artifact contract.

Component	Value	Component	Value
Map size	96×96	Exploration budget	$T_{\text{exp}} = 800$
Motifs	24	Splits	10 train, 6 val, 8 test
Families	metric/topology/dynamics/semantic	Severities	0, 1, 2, 3
Task classes	planning/inference/adaptation	Task samples	3 per class/cell
Trace-trained seeds	5 per learned baseline	Bootstrap	1000 grouped resamples

Table A.4: Protocol sensitivity for mixed family-primary scores. Ranking comparisons are computed from the same generated 24-motif study grid while changing one evaluation assumption.

Comparison	Kendall τ	Rank reversals	Family changes
No exploration vs. reward-free exploration	1.000	0	1
Same-environment vs. CPE	1.000	0	2

This table uses mixed family-primary scores aggregated over planning, inference, and adaptation; task-conditioned planning slices are analyzed separately in Section 8 and Table A.13.

Table A.5: Motif-wise descriptive consistency for the mechanism diagnostic. Entries average per-motif belief-update minus stale-map scores equally over eight test motifs; Table 2 reports the primary paired grouped estimate and interval. The final column counts CPE wins and motifs in which CPE enlarges the update advantage.

Family	Motifs	Mean CPE gap	Mean same-env. gap	Mean protocol delta	BU wins / reduction
Topology	8	0.336	0.102	0.234	8/8; 7/8
Semantic	8	0.724	0.102	0.622	8/8; 8/8

Table A.6: Severity-response summary from the full run. Entries are family primary scores for two calibration baselines across severities 0–3. The severity health gate validates monotone intervention magnitudes, not monotone empirical scores; full method-by-family severity data and rendered curves are included in the artifact.

Baseline	Family	S0	S1	S2	S3
Privileged planning ref.	Metric	0.521	0.486	0.471	0.443
Privileged planning ref.	Topology	0.521	0.552	0.750	0.740
Privileged planning ref.	Dynamics	0.521	0.760	0.792	0.781
Privileged planning ref.	Semantic	0.521	0.802	0.833	0.823
Weak heuristic	Metric	0.512	0.440	0.409	0.407
Weak heuristic	Topology	0.486	0.566	0.444	0.516
Weak heuristic	Dynamics	0.512	0.677	0.697	0.693
Weak heuristic	Semantic	0.486	0.122	0.122	0.141

Table A.7: Family-wise CPE calibration results on non-identity severities 1–3. Entries are mixed primary family scores with equal severity and task-class weights, not planning-reference-normalized upper-bound scores. Deterministic/reference rows use the expanded 24-motif mechanism diagnostic. Learned rows are capacity-calibration rows and are not used for the held-out stale-map versus belief-update claim. The weak heuristic is a hand-coded floor with local spatial priors; full 95% grouped-bootstrap confidence intervals are included in the artifact JSON and rendered Markdown tables.

Method	Metric	Topology	Dynamics	Semantic
Privileged planning ref.	0.439	0.698	0.656	0.761
Weak heuristic	0.375	0.489	0.488	0.167
Classical belief update	0.485	0.689	0.648	0.810
Pretrained graph world model	0.295	0.327	0.400	0.401
Persistent memory	0.313	0.332	0.323	0.354
Relational graph	0.252	0.261	0.252	0.204
Structured dynamics	0.259	0.242	0.280	0.218

Table A.8: Family-wise mechanism diagnostic scores under CPE on non-identity severities 1–3. Scores are mixed primary family scores with equal severity and task-class weights. Planning-only reference path length and reference-gap metrics are tracked separately in the artifact. The diagnostic separates stale pre-intervention reuse, local heuristic priors, and explicit belief update.

Method	Metric	Topology	Dynamics	Semantic
Privileged planning ref.	0.439	0.698	0.656	0.761
Stale-map planner	0.424	0.395	0.656	0.178
Weak heuristic	0.375	0.489	0.488	0.167
Classical belief update	0.485	0.689	0.648	0.810

B Baseline metadata

Table A.9: Deterministic reference baselines. These methods have no trainable parameters and are included to calibrate solvability, protocol sensitivity, and weak-reference behavior.

Baseline	Access and stored state	Evaluation behavior	Training
Privileged post-intervention planning reference	Privileged access to the evaluation environment for solvability and planning-reference checks.	Deterministic shortest-path planning, exact inference answers, and reference adaptation curve when solvable.	0 params; none.
Same-environment reference condition	Protocol-control wrapper used only when execution is performed in the explored environment rather than the intervened environment.	Shares deterministic planning mechanics with the corresponding evaluation condition; interpreted as a protocol comparison condition.	0 params; none.
Weak heuristic	Stores visited cells, visited node ids, visible goal-token bindings, and limited local traversable structure. No global map revision.	Orthogonal visible-path planner over remembered structure; always predicts no topology change; answers semantic reachability with stale remembered token bindings; predicts masked family by task-type rule; capped heuristic adaptation curve.	0 params; none.
Stale-map planner	Stores the serialized pre-intervention environment from reward-free exploration and deliberately does not update it after intervention.	Plans shortest paths and answers semantic queries using the pre-intervention map, then is scored in the evaluation environment. Used as a diagnostic sensitivity floor for stale exploration-derived state.	0 params; none.
Classical belief update	Stores an occupancy map, edge structure, dynamics signature, and semantic-token bindings from exploration.	Uses local post-shift mismatch observations and the task/adaptation budget to update edges, costs, dynamics, and token bindings, then replans. It does not receive the intervention family label or global post-intervention map.	0 params; none.

Training source. Persistent-memory, relational-graph, and structured-dynamics learned baselines are trained separately for each base environment and model seed from the current exploration instance. The pretrained graph world model is a larger capacity-control run trained on generated graph instances before CPE evaluation. In all cases, evaluation-time tensors are built from the exploration-derived graph: node features contain normalized position, local free/blocked/semantic observation statistics, role indicators, and global dynamics features; pair labels supervise edge existence, normalized geometric path cost, and normalized traversal cost; node labels supervise semantic-token location.

Optimization and hyperparameters. Relational and structured-dynamics learned baselines use Adam with learning rate 0.01, validation fraction 0.2, and early-stopping patience 3. The large persistent-memory run uses 2048 memory slots, readout width 672, Adam learning rate 0.0005, 120 epochs, and patience 20. The pretrained graph world model uses Adam with learning rate 0.0005, 120 epochs, and patience 20. Hyperparameters are specified in configuration files before final evaluation; validation motifs are the only intended place for pre-final configuration choices. Test motifs are not used for hyperparameter selection. Checkpoints are keyed by baseline name, environment id, model seed, and hyperparameter hash, then reused across protocol variants.

Compute reporting. The smoke path is designed to run on CPU and completes in approximately 2–5 minutes for the smoke build itself; the one-command artifact audit takes longer because it also runs the test suite. The expanded 24-motif full study is intended for a single CUDA GPU; on an NVIDIA L4 GPU with 23GB memory, reviewers should budget roughly 30–36 wall-clock hours, with faster completion expected on L40S/H100-class GPUs. Preliminary/debug runs are excluded

from the main reported compute. The learned training jobs are small; wall time is dominated by task generation, planning/evaluation loops, bootstrap aggregation, and artifact rendering.

Table A.10: Learned baseline metadata. Parameter counts are representative realized trainable parameters for the release configuration; exact per-run counts, device metadata, checkpoints, and training summaries are written to run manifests.

Baseline		Stored representation and architecture	Supervision and optimization	Size
Pretrained world model	graph	Larger graph message-passing model with hidden size 256, 6 message-passing steps, pair width 256, and dynamics width 128.	Pretrained on 4000 generated training graphs with 800 validation graphs, then evaluated under the same CPE task grid; 120 epochs; Adam lr 0.0005; patience 20; 2592 non-identity evaluation episodes.	~1.14M params.
Persistent-memory world model		Builds the same graph data, but reads each node through learned memory slots. Slot attention stores persistent exploration-derived state.	Large run with 2048 memory slots and readout width 672; same self-supervised heads; 120 epochs; Adam lr 0.0005; validation fraction 0.2; patience 20; 2592 non-identity evaluation episodes.	~1.4M params.
Relational world model	graph	Uses node features and the explored adjacency matrix; applies a node encoder plus 2 message-passing steps over normalized adjacency.	Same self-supervised targets; 10 epochs; Adam lr 0.01; validation fraction 0.2; patience 3; 5 seeds.	396 params.
Structured-dynamics world model		Factorized geometry and dynamics encoders; edge and geometry heads use geometry factors, while traversal-cost prediction additionally receives global dynamics factors.	Same self-supervised targets; 10 epochs; Adam lr 0.01; validation fraction 0.2; patience 3; 5 seeds. Structured by restricting which latent factors feed each prediction head.	504 params.

C MiniGrid adapter contract

This appendix makes the substrate-agnostic claim operational. The artifact includes a MiniGrid-compatible adapter that keeps the CPE interface fixed while replacing the current map generator with a MiniGrid-style environment wrapper. The adapter implements deterministic base sampling, declared intervention application, pair validation, optional instantiation of an actual `minigrid` environment, and a smoke command. The output schema follows MapShift: split id, base environment id, intervention family, severity, rejection reason if any, reference-solvability status, and validator records. The point is not to make MiniGrid look like MapShift, but to make a richer substrate satisfy the same measurement contract.

Table A.11: MiniGrid adapter contract. The included adapter implements the symbolic-grid column; visual or language-conditioned wrappers would add the redesign checks in the right column.

Component	MiniGrid instantiation	Validators that transfer	Validators needing re-design
Base generator	Seeded rooms, doors, keys, walls, lava, and goal objects; motif id replaces map motif.	Split leakage, motif counts, reachable-state coverage, reference solvability.	Mission-template leakage if natural-language missions are used.
Metric shift	Change action cost, step budget, turn/forward cost, or grid-edge weights without changing walls or object identities.	Topology preservation, object identity preservation, path-cost monotonicity.	Cost semantics must be documented because MiniGrid has discrete actions rather than metric distance.
Topology shift	Add/remove doors, walls, blockers, or one-way passages while preserving object identities and start/goal distribution.	Connectivity delta, non-target semantic invariance, reference solvability, task rejection counts.	Door/key dependencies require checking both geometric and inventory-conditioned reachability.
Dynamics shift	Introduce slip, stochastic action failure, locked-door transition changes, or lava penalty changes.	Geometry and semantic preservation, severity-response checks, weak-baseline non-saturation.	Stochastic transitions require repeated rollouts or analytic transition matrices for reference scoring.
Semantic shift	Remap goal color/object type, key-door binding, or mission target while preserving geometry.	Geometry/topology preservation, target-binding change, non-vacuous task checks.	Language templates must be checked so the answer is not leaked by wording.
Tasks	Post-shift navigation, changed-door/key inference, masked-cell prediction, and limited interaction followed by replanning.	Family/task coverage, horizon distributions, final-release rejection accounting.	Observation wrappers must specify whether agents receive symbolic grids, pixels, missions, or both.

Table A.12: MiniGrid-compatible sanity-study result. This is a symbolic MiniGrid-style grid wrapper with optional `minigrad` instantiation, not a pixel/VLA evaluation. The same deterministic mechanism diagnostic is run as in Section 8: privileged planning reference, stale-map planner, weak local planner, and classical belief update under CPE, same-environment, and no-exploration controls.

Quantity	Result
Grid and splits	6 MiniGrid-style motifs, 24 environments; train/val/test = 12/4/8.
Matched intervention pairs	384 pairs across metric, topology, dynamics, and semantic families.
Health checks	0 task rejections, 0 validation failures, reference solvability 1.000.
Weak baseline	Weak local planner score 0.496; non-saturating.
Protocol-sensitive gaps	Belief-update minus stale-map gap under CPE is 0.313 for topology and 0.750 for semantic shifts; same-environment and no-exploration gaps are 0.000 in those controls.
Ranking effect	Semantic same-environment vs. CPE has Kendall $\tau = 0.667$ with one rank reversal.

The same acceptance gates apply before model interpretation: leakage and validator failures must be absent, rejection reasons reported by cell, reference tasks solvable, and the weak baseline non-saturating. The implemented ProcTHOR study replaces grid-graph checks with certification of realized object and door edits, navigable geometry, visibility, and feasibility. A further simulator or hardware port needs its own equivalent invariants. The MiniGrid adapter remains a compact test of the interface and validators.

D Conclusion audit across studies

The audit below collects the interpretation of the reference, MiniGrid, and ProcTHOR findings. The symbolic MiniGrid adapter’s distinct sanity-study values remain in Table A.12; the held-out 3D study supplies the richer observation and learned-system tests. These conclusions concern specific comparisons under matched changes, not every aggregate leaderboard.

Table A.13: Conclusion audit across the reference and held-out 3D studies. The source column identifies the evidence for each protocol-dependent interpretation.

Standard conclusion	Source	MapShift diagnosis	Why it matters
Same-env task ranking is enough.	24-motif mechanism diagnostic	Planning slices show protocol-induced reversals: metric/dynamics have $\tau = 0.333$ with two reversals; topology severity-1 planning has $\tau = 0.000$ with three.	Standard protocols can change method order.
Stale maps are competitive.	Mechanism diagnostic	Belief update beats stale-map reuse on all 8 held-out topology and semantic motifs under CPE.	Updating, not reuse, is the target competence.
Memory capacity is enough.	Capacity runs	Large persistent memory improves to 0.313–0.354 but remains below update-based references on most families.	Capacity does not substitute for explicit update.
Same-env gaps estimate the intervention effect.	Paired deltas	CPE increases the belief-update advantage over stale-map by 0.201 on topology and 0.622 on semantic shifts.	Protocol assumptions change conclusions.
A high unchanged-house score establishes post-change repair.	371 ProcTHOR houses	Semantic relocation exposes a +0.532 local-update success gap that rounds to zero in the unchanged control.	Post-change evidence must be tested.
The unchanged-house choice of belief interface transfers.	Paired RSSM systems	Grounded full repair leads by 0.041 unchanged; RSSM-only full repair leads by 0.224 after relocation.	Evaluation can reverse system choice.

E ProcTHOR effect-size visualization

F Artifact contents

The release artifact is an executable benchmark and generator. It contains code, versioned configs, schemas, and commands to regenerate split manifests, interventions, task records, health reports, family-wise tables, severity responses, protocol comparisons, and provenance. The README identifies the paths below; the ProcTHOR study’s pair manifests, rejection ledger, and house-clustered records support offline recomputation. Any separately hosted generated dataset should carry Croissant core and Responsible-AI metadata.

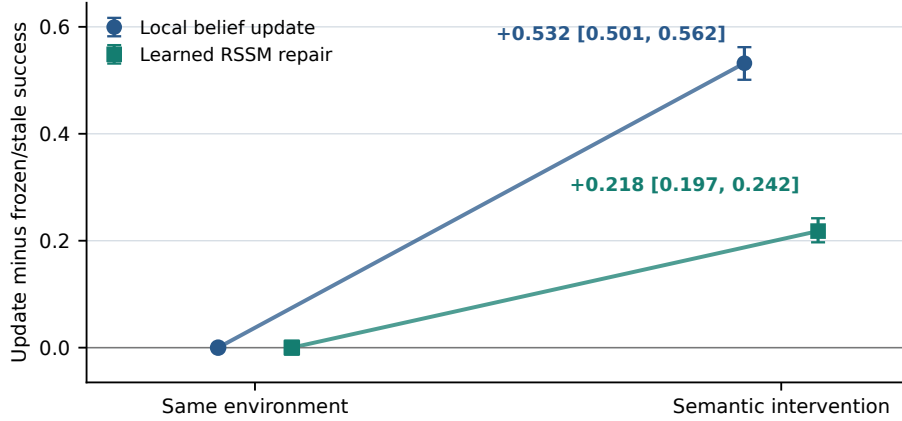


Figure A.1: Paired semantic-navigation success gaps in 371 contributing ProcTHOR houses. Unchanged-world gaps round to zero; CPE distinguishes stale reuse from local update and grounded frozen from grounded full-repair RSSM beliefs. Brackets show 95% house-bootstrap intervals.

Table A.14: Reference-artifact reproduction routes. The full-study runtime refers to a new run; tables and figures can be regenerated in minutes from existing records.

Route	Expected output
CPU smoke (2–5 min)	Small records, configs, and sanity tables
Health audit	Leakage, rejection, validator, solvability, and coverage report
Result reproduction	Table 2, Figure 3, intervals, and assets; full 24-motif rerun takes roughly 30–36 h on an L4 GPU
MiniGrid smoke	24 environments, 384 pairs, health checks, and solvability 1.000

G Broader impact and ethics

MapShift is an evaluation artifact for embodied-planning research. It does not contain human-subject data, scraped web data, personally identifiable information, or deployed decision systems. The main risk is misuse of benchmark scores as evidence for real-world autonomy or robotics safety beyond the benchmark assumptions. The paper mitigates this by stating supported and unsupported claims, reporting benchmark health separately from model results, and emphasizing family-wise diagnostics over a single leaderboard score.

H NeurIPS Paper Checklist

1. **Claims.** Yes. The abstract and introduction state that MapShift is an executable post-intervention evaluation protocol and benchmark. The empirical claims concern the 24-motif reference study and held-out ProcTHOR navigation under the stated observation and planning interfaces.
2. **Limitations.** Yes. Section 10 states the supported scope and limitations for interpreting benchmark results.
3. **Theory, assumptions, and proofs.** N/A. The paper formalizes an evaluation protocol and estimands but does not present theoretical theorems requiring proof.
4. **Experimental result reproducibility.** Yes. Methodology, artifact, and appendix sections describe versioned configs, task generation, baseline training, and grouped bootstrap estimation. Section 7 states the held-out house split, certification counts, observation interface, and model-seed design for the 3D study.
5. **Open access to data and code.** Yes. The submission includes an executable artifact containing benchmark code, versioned configs, schemas, and reproduction commands for the reference study; Section 7 describes the 3D study and its reported evaluation contract.
6. **Experimental setting/details.** Yes. The paper specifies splits, exploration budget, intervention families and severities, task classes, baseline mechanisms, model seeds, certification, and aggregation, including the ProcTHOR house-level design.
7. **Experiment statistical significance.** Yes. The reference study reports grouped-bootstrap intervals; the ProcTHOR study reports paired 95% intervals clustered by held-out house, with learned-system seeds paired within house.
8. **Experiments compute resources.** Yes. Appendix B reports the expanded full-study setting: a single CUDA GPU, with roughly 30–36 wall-clock hours budgeted on an NVIDIA L4 GPU. It also reports the CPU smoke-test setting and clarifies that preliminary/debug runs are not included in the main reported compute.
9. **Code of ethics.** Yes. The work is designed as a controlled evaluation artifact and does not involve human-subject data, private data, or high-risk deployed systems.
10. **Broader impacts.** Yes. The appendix discusses the main risk: over-interpreting benchmark performance as real-world autonomy readiness.
11. **Safeguards.** N/A. The paper does not release a high-risk pretrained model or dual-use generative model.
12. **Licenses.** Yes. The released code is licensed under the MIT License; generated benchmark outputs and rendered result artifacts are licensed under CC BY 4.0; third-party dependencies and referenced tooling are listed in `THIRD_PARTY_LICENSES.md`.
13. **Assets.** Yes. The paper documents the released executable benchmark artifact, its output schema, limitations, and intended use. No static dataset is introduced; if generated outputs are hosted as dataset-like assets, they should be accompanied by Croissant/RAI metadata.
14. **Crowdsourcing and human subjects.** N/A. The work uses no crowdsourcing and no human subjects.
15. **IRB approvals.** N/A. No human-subject research is conducted.
16. **Declaration of LLM usage.** N/A. LLMs are not a component of the benchmark methodology, baselines, or experimental results.